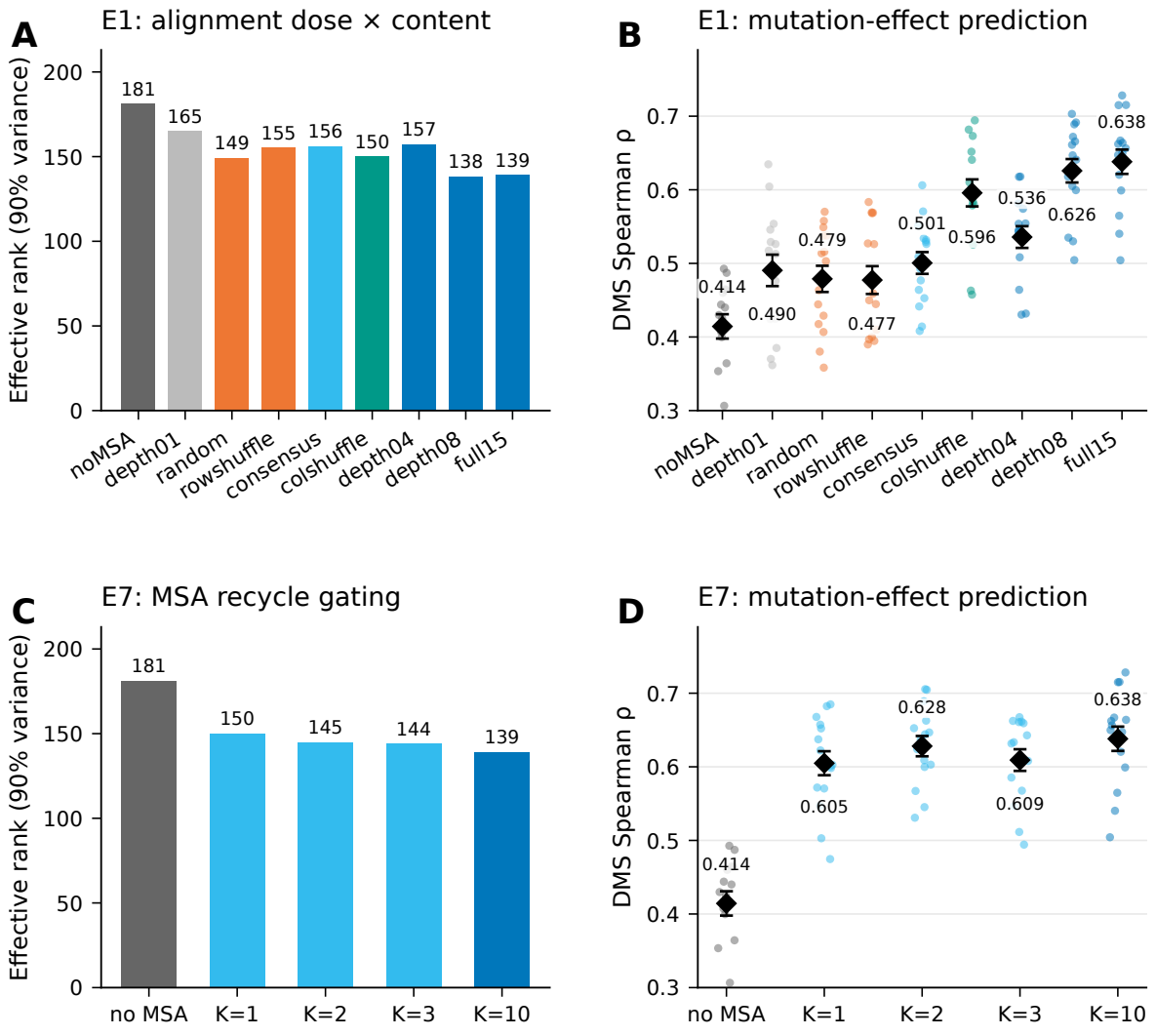

A
E1: alignment dose × content
Effective rank (90% variance)
181
165
149
155
156
150
157
138
139
noMSA
depth01
random
rowshuffle
consensus
colshuffle
depth04
depth08
full15
B
E1: mutation-effect prediction
DMS Spearman ρ
0.414
0.490
0.479
0.477
0.501
0.596
0.536
0.626
0.638
C
E7: MSA recycle gating
181
150
145
144
139
no MSA
K=1
K=2
K=3
K=10
D
E7: mutation-effect prediction
0.414
0.605
0.628
0.609
0.638